



From Document to Meaningful Exploration (in progress): *Visualising Cybersyn Documents*

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University of Edinburgh (Professor: Melissa Terras & John Vines)

SB100Y Conference
18, Sept, 2026

Outline

Two main parts

a- Personal perspective

b- Project

Personal Relation to Technology

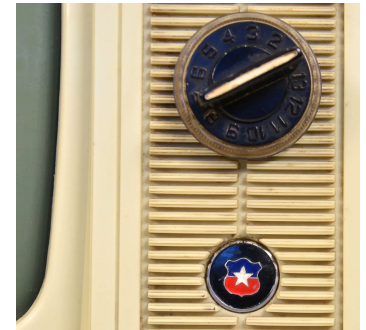
Where I come from...

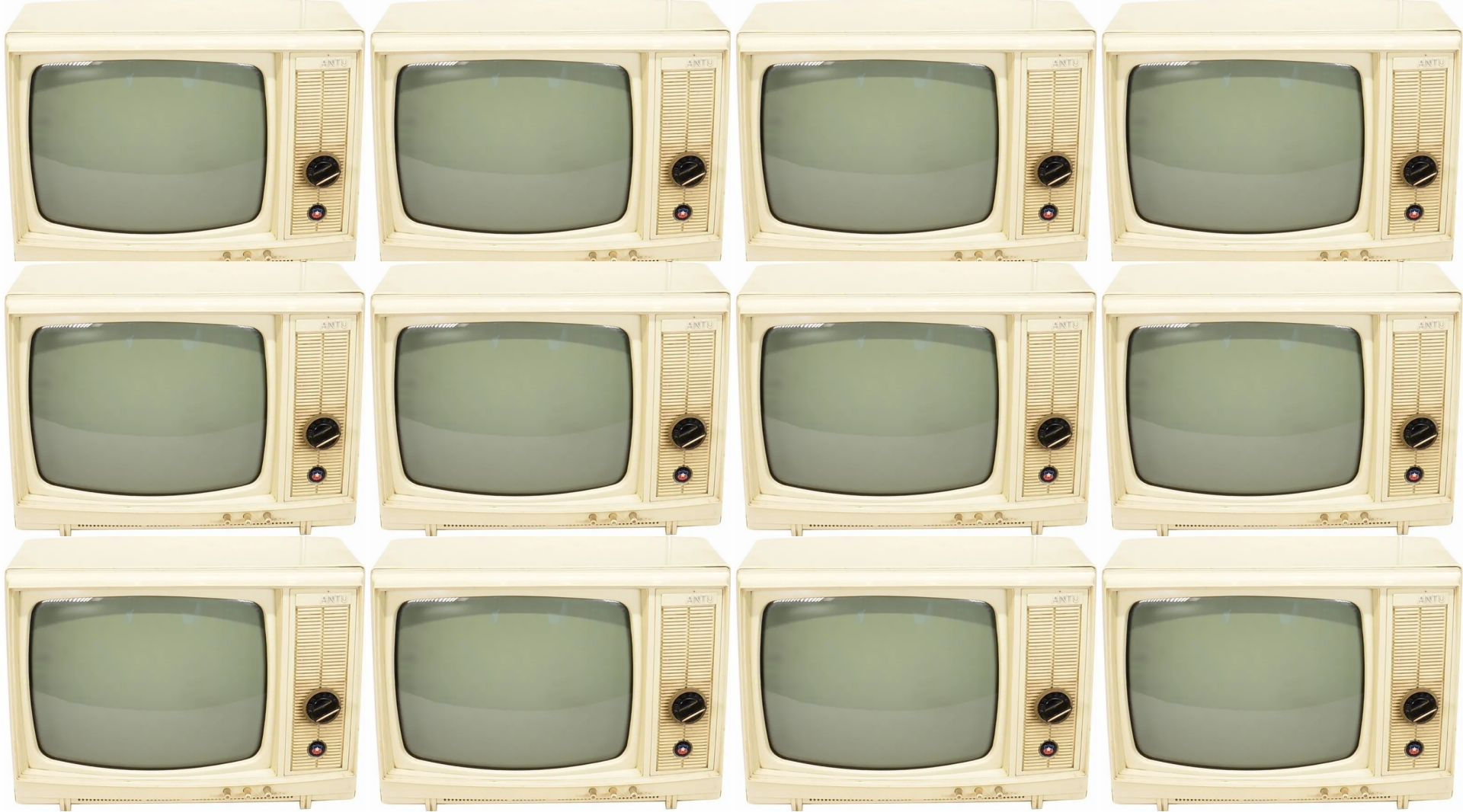


Santiago, Chile.
Designer, artist, media artist.

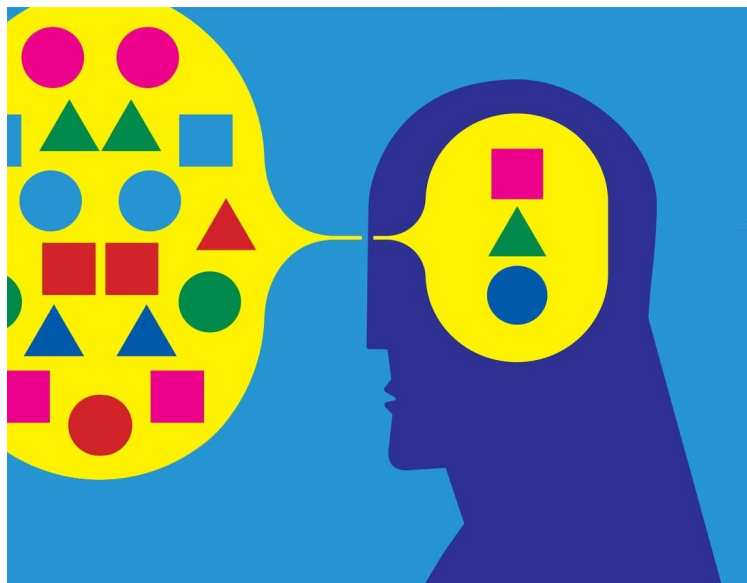
Childhood during the dictatorship, yet surrounded by technology:
televisions (Antu), radios, irons, etc. Electricity.

ANTU TV made in Chile
in the 70's





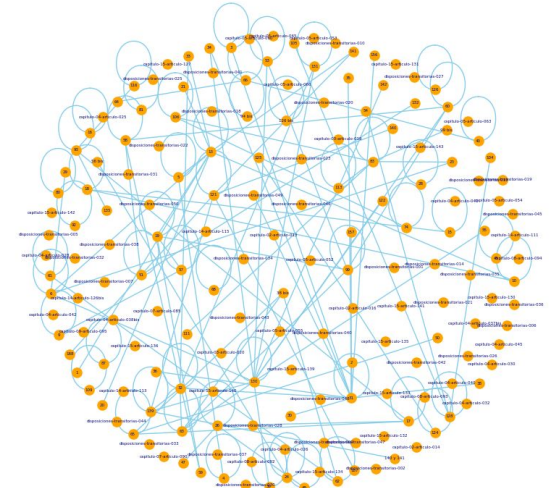
Interests in technology and creation: Visual explorations



6

drawing-design | programming | text | culture | language as technology | Code & art | technologies and politics

Do you what the Constitution says about indigenous people?															
Answers? explore: Environment, Parliamentary rules, Politics and power, Privacy, Intellectual property															
Articles n°: 4, 19, 116															
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Cross citation articles analysis

Disciplinary Interests In Cybersyn (Chile)

Movement to recover the history of Latin American technology and Cybernetics:

Dusan Cotoras (sociologist), Rodrigo Fernández (sociologist), David Maulen (art researcher), Hugo Palmarola (designer) & Pedro Alonso (architect), Fernando Portal (architect), Enrique Rivera & Catalina Ossa (artists), Leonardo Portus (artist), Claudio Gutierrez (computing) & Juan Álvarez (computing), Juan Carlos Letelier (biologist), etc.

Some question remains:

What other issues and figures require research? Arturo Aldunate Philips, Ricardo Uribe.



Chilean Scientist Arturo Aldunate Philips & Norbert Wiener (nd).

Does cybernetics persist in Chile?



Instituto de filosofía en Ciencias de la complejidad, Dirección Pablo Razeto.

Disciplinary Interests In Cybersyn

Initial encounters with the figure of Stafford Beer:

Article on Cybersyn (Enrique Rivera & Catalina Ossa, artists):

–Colectivo Troyano, Ignacio Nieto, Italo Tello, Ricardo Vega (Art):

Book: Instalando: Art & Technology (2006)



LAS HERRAMIENTAS DE LA CIENCIA PARA EL PUEBLO.

Catalina Ossa y Enrique Rivera

Cuando se planeaba el lanzamiento oficial de Cybersyn a la sociedad chilena, en una fecha ya confirmada para Septiembre de 1973, el Presidente Salvador Allende preparó un especial discurso que pronunciaría en la inauguración.

"Estimados amigos, quiero darles la bienvenida personalmente a este lugar, ya que he tenido un interés personal en su desarrollo, y por que quiero pedirles un especial interés en él. Lo que ven alrededor es el resultado de 18 meses de trabajo duro por un grupo sólido de nuestros propios ingenieros chilenos, que se han dedicado a solucionar los problemas de la gestión en las empresas. Ellos han creado para nosotros, una serie de herramientas para ayudarnos en la tarea de controlar la economía.

La ciencia moderna, y en particular la computación electrónica, ofrecen al gobierno una nueva oportunidad para tratar con los complejos problemas modernos de la economía.

Hemos encontrado que en los llamados países avanzados, el poder de la ciencia no ha sido utilizado aún. Hemos desarrollado un sistema con nuestro propio espíritu. Lo que escucharan hoy día es revolucionario –no simplemente por que esto es la primera vez que esto se realiza en el mundo. Es revolucionario por que estamos ante un esfuerzo deliberado para darle a la gente el poder que la ciencia nos da, en una forma en la cual la gente podrá usarla libremente."

Lamentablemente, este discurso nunca pudo ser pronunciado debido a la desestabilización social y económica del Gobierno Socialista, causada por factores internos y externos, que finalmente terminaron con el Golpe Militar de Septiembre de 1973.

Las herramientas a las que se refiere Allende se podrían identificar hoy como los primeros intentos de una emergente realidad sociocultural, que hoy en día son fáciles y utilizadas gracias a diversos avances tecnológicos, y a las posibilidades de interacción a distancia que permite INTERNET, enmarcados en los paradigmas propuestos en la era de la Sociedad de la Información y la emergente Sociedad del Conocimiento.

La revolución pacífica, económica, social y tecnológica que representaría Chile estaría materializada por el proyecto CYBERSYN.

Disciplinary Interests In Cybersyn

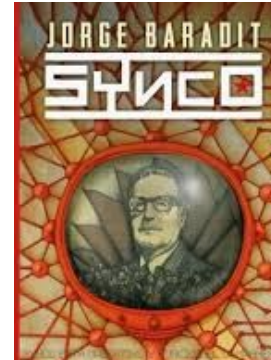
What is the relationship between art and technology?
Exploratory art and design projects regarding Cybersyn:



–Enrique Rivera & Catalina Ossa (2002, artists): thesis project at ZKM | Center for Art and Media, Karlsruhe.



–Greg Borenstein & Jem Axelrod (2006, media artists): critical explorations of the interface from a science fiction perspective.



–Jorge Baradit (2006, writer), "SYNCO".



–Mario Navarro (2006, artist): video and installations concerning Stafford Beer "Whisky in OpsRoom"

Disciplinary Interests In Cybersyn

What is the relationship between art and technology? Exploratory art and design projects regarding Cybersyn:



–Diego Gómez (2014, designer):
archive, documentation, and references



–Luke Munn & Liam Magee (2024,
media & social sciences): Rethink
Economic Futures AI interviews using
LLMs.



–Hugo Palmarola, Pedro Alonso, Edén
Medina (2025, designer, artist,
historian): exhibition and book, real size
OpsRoom.

Visualisation Project

Questions I had / Needs observed

data processing
& analysis

difficult
complex
time consuming
...

—

data representation
& communication

fit data with graphics
standard graphics not appropriate
time consuming
...

The process was slow, heavy, technically difficult for non-experts...

so...

How to optimize tasks within the process?

How to understand and manage the entire process?

Fit of interests

Personal
interest

+

Document processing
for Humanities

+

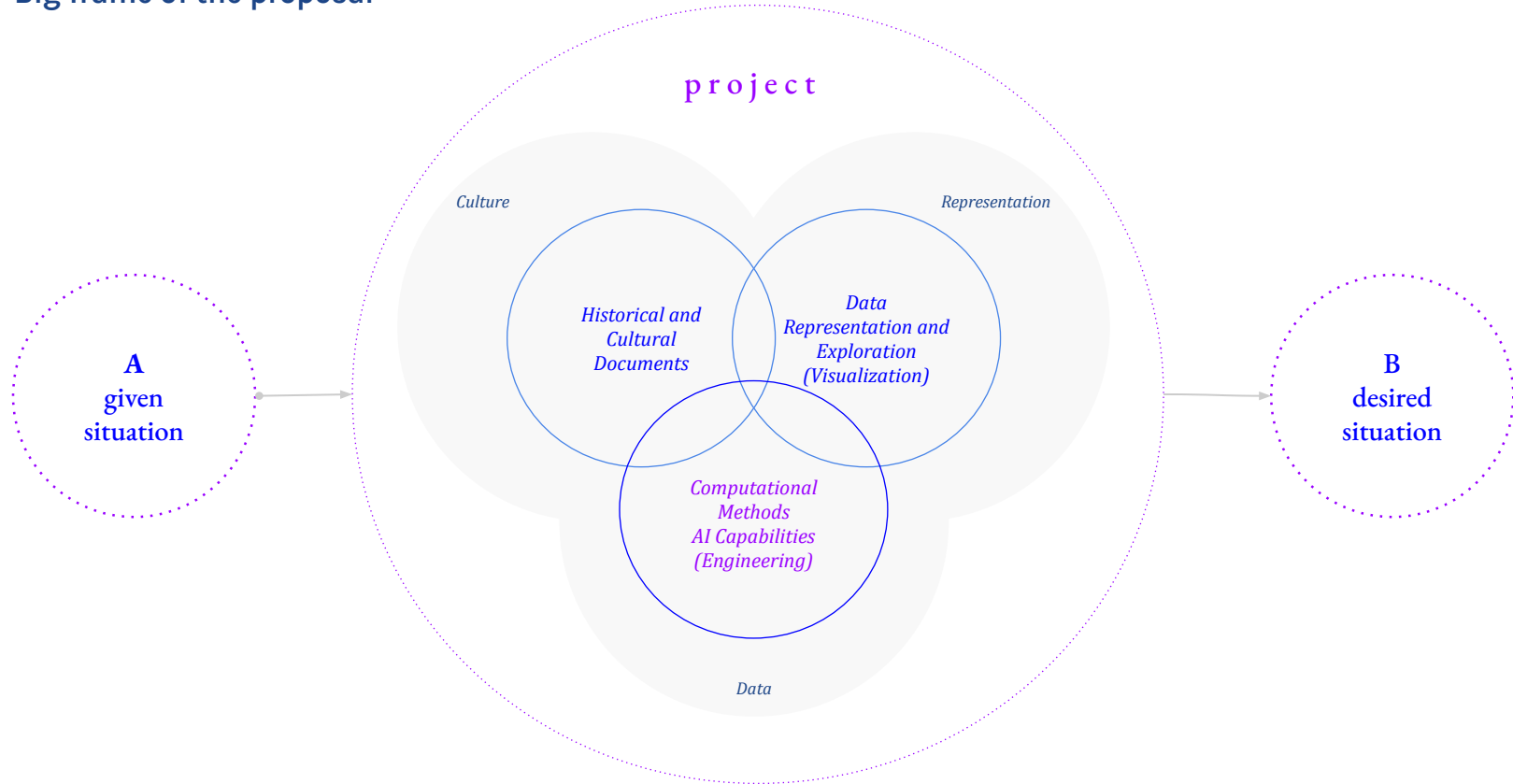
Apply AI
to Visualization

visualization for cultural documents
& information asymmetries
(Weaver, 1948; Maeda, 2013)

understanding the data
& visualization process
(Hinrichs et al., 2015)

facilitating data processing
& visualization tasks
(Basole et al 2024; Wu et al., 2021;
Wang et al., 2023)

Big frame of the proposal



Proposal

The proposal is to develop an **interactive exploration system** that, through **artificial intelligence-assisted** queries to the database, enables the **generation of interactive visualization narratives** about the relationships and stories of **people, places, or events**.

Relevance & Research Contribution

Visualization of cultural Documents:

This is an opportunity to access the documents and facilitate its visual exploration
(Hinrichs et al., 2015)

Cultural Analytics for the Archive:

The cultural analytics framework will provide another perspective for understanding documents
(Schich et al. 2014; Aiello & Simeone, 2019)

Prototype of Framework for Visualization:

This Framework/Process may serve as a guide for implementing other case studies with multimodal cultural documents.
(Dürsteler & Engelhardt, 2007; Basole et al 2024)

Disciplinary Integration:

This an opportunity for disciplinary integration
(Weaver, 1948)

Recovery of Technological Heritage:

Relevant for understanding current technological developments. Cybersyn is a case study
(Letelier, 2022; Medina, 2014)

Process Overview:

To understand how LLMs and AI expand the creation and analytical possibilities of the visualization process (Dürsteler & Engelhardt, 2007; Basole et al 2024)

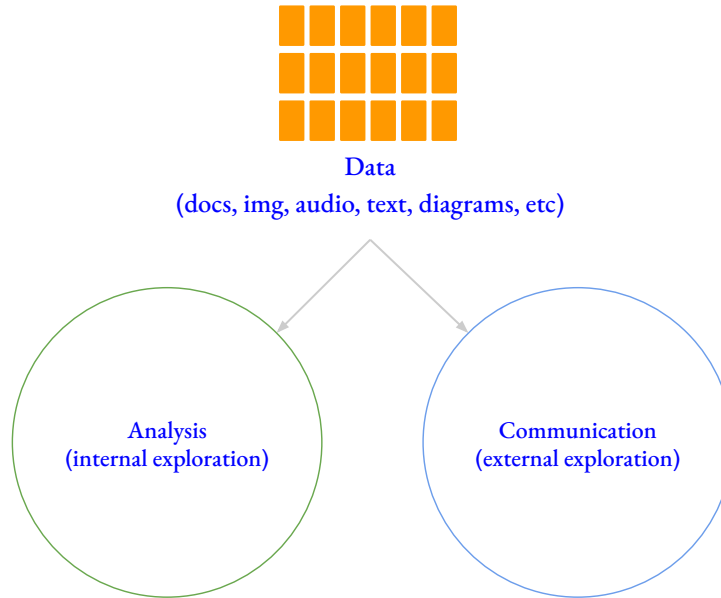
Theoretical Framework

Visualization

Reduces complexity using visual elements

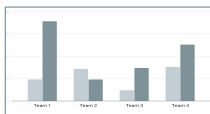


Visualization Main Roles



Visualization

The broader visualization spectrum...



Pragmatic
Visualización



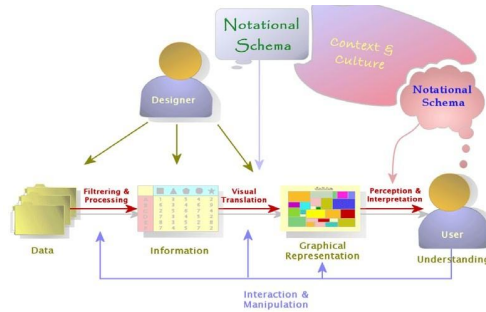
Data Visualization - Exploratory Visualization - Casual Infovis - Physical Visualization - Environmental Visualization - Data Art



Artistic
Visualization

Visualization

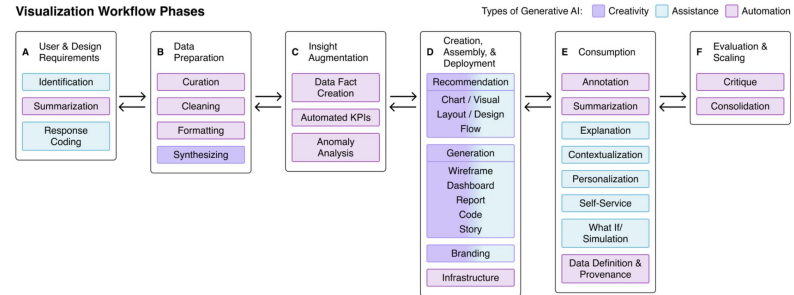
Models: How to understand visualization tasks



"Infovis Diagram":

Most general description of the visualization process, but not considers the AI development.

Visualization Workflow Phases

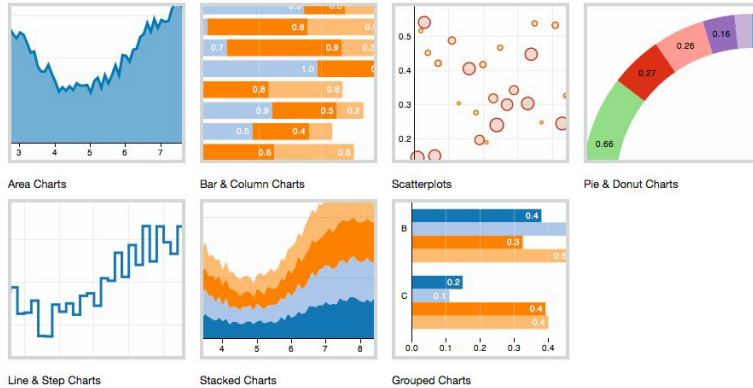


"Visualization Workflow Phases":

Recently discovered, more detailed, focused in find how to integrate AI.

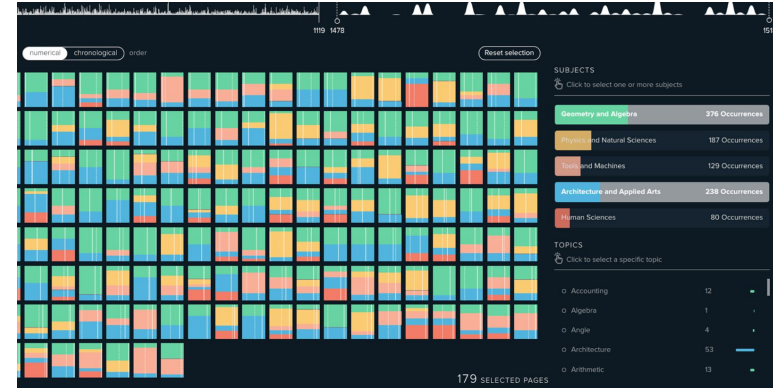
Visualization

Standard graphics vs custom made



Standardized libraries,
Graphics “templates” for each kind of visualization.
Matplotlib, Seaborn (Python)
D3, Protovis (Javascript)
GGPLOT (R)

- limited customization
- mainly for analytics
- + well suited for data
- ...



Custom made visualization
From the ground or based on previous libraries.

- + control over output
- + control over visual style
- + adapted for specific cases
- ...

Visualization

Custom visualization



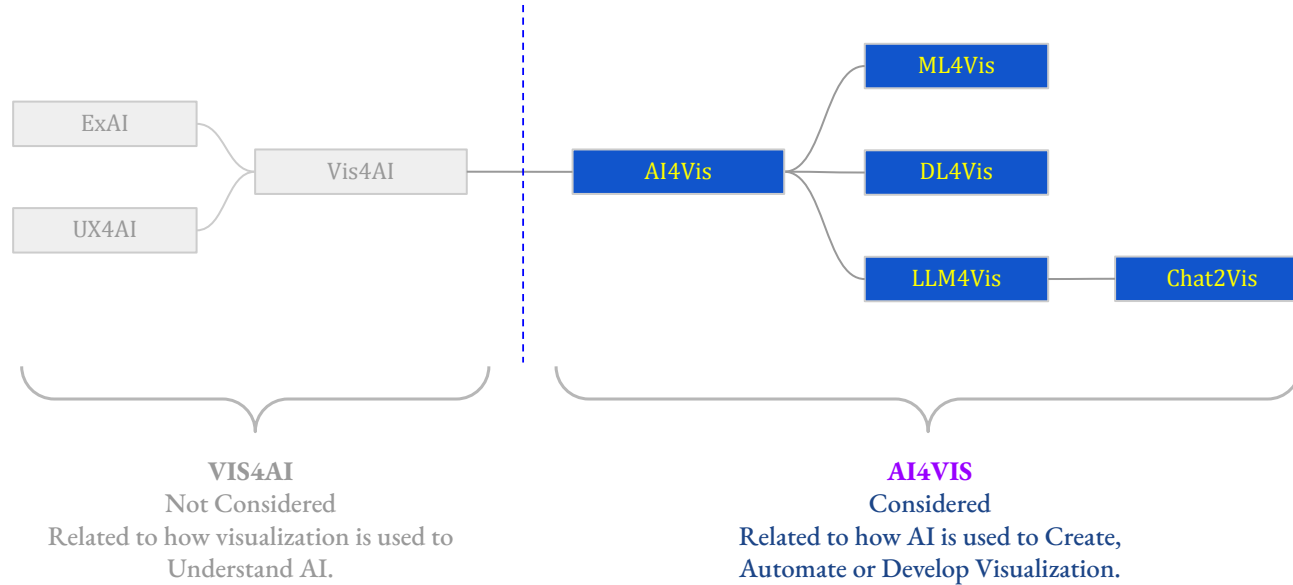
Codex Atlanticus

Interactive website, allows you to navigate through pages, sketches and notes from Leonardo da Vinci Codex Atlanticus.

link: <http://codex-atlanticus.it/#/?lang=EN>

State of the Art

AI and Visualization current research - Two sides



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Study Case

Case Study: Stafford Beer Archive, Liverpool John Moores University (LJMU) Documents of relevant project, Chile 1972.

Then



Stafford Beer
English scientist.

Ops Room and usage of
visualizations

Now



Eden Medina Book

Recent render reconstructing
the Ops room.

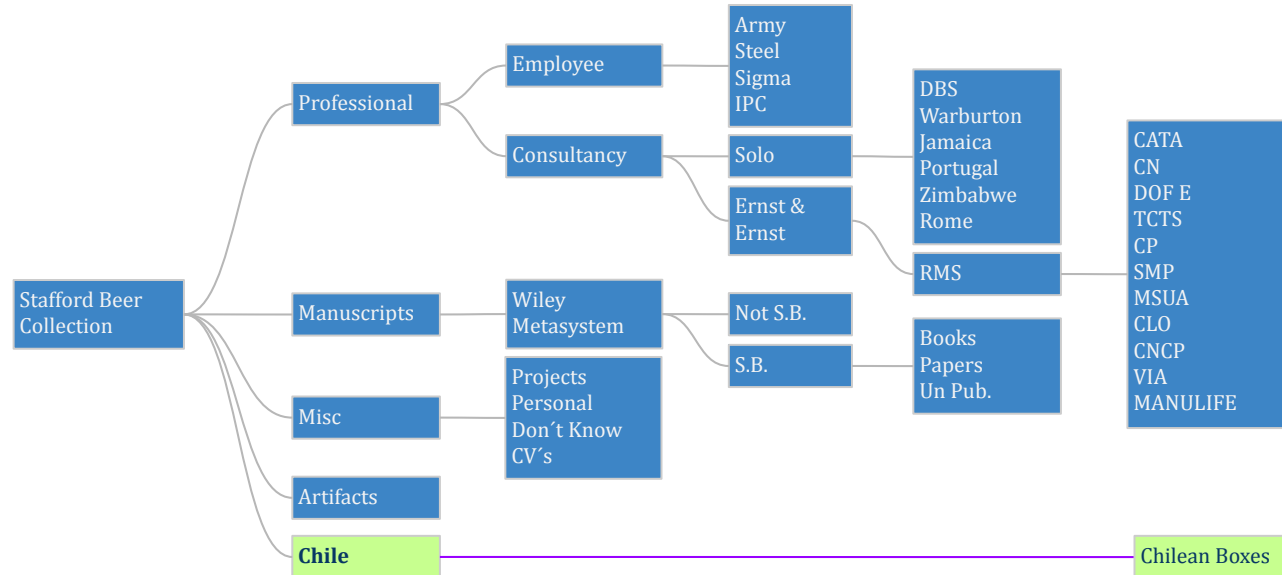
Recent Exhibition
Book

Stafford Beer Archive, Liverpool John Moores University
Special Collections & Archives.

**There is no visual access to explore the
documents on which the research is based.**

Case Study

SBA (LJMU) - Structure



Case Study

SBA (LJMU) - Available Boxes

Stafford Beer
Collection

Chilean Boxes

Box 55 (10 documents): "Hello," and "Please stay". Some items are on loan from this box
Box 56 (3 documents): The Big Black File. This whole box is on loan, and not currently accessible
Box 57 (8 documents): Cyber Stride. Some items are on loan from this box
Box 58 (3 documents): Ron Anderton
Box 59 (1 documents): Arthur Andersen
Box 60 (4 documents): Cybersyn. Some items are on loan from this box
Box 61 (5 documents): Project Cyberfolk. Some items are on loan from this box
Box 62 (15 documents): Ops Room. Some items are on loan from this box
Box 63 (15 documents): "Snapshots". Some items are on loan from this box
Box 64: (15 documents): Corfo Miscellaneous.

Boxes
Available

Some items are on loan from this box. Checked up to here 12/05/2022

Box 65: Corfo Miscellanea 2
Box 66: Correspondence
Box 67: Films, Accounts
Box 69: Chile -POST COUP -Relief
Box 70: POST-COUP Correspondence FOR THE ATTENTION OF SB BEER
Box 71: Related Journals

Boxes Not
Available

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Cybersyn Archives Related to Chile

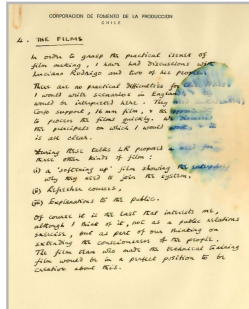


Documentos por cada caja

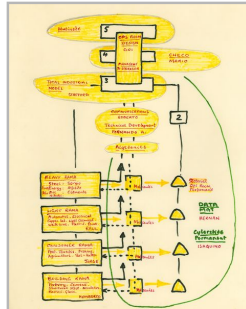
Box 55	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400
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Case Study

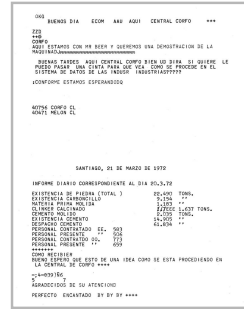
SBA (LJMU) - File Types



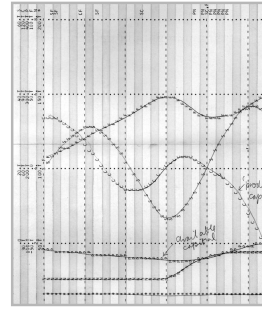
—manuscript (152)



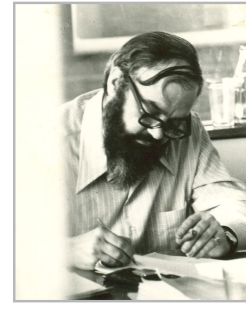
—manuscript diagram
(138)



—official document (74)



—diagram (31)



—photo (15)

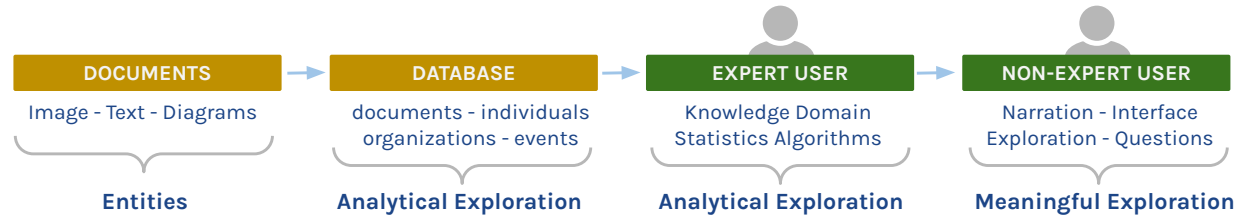


—official letter (8)

Methodology

Methodology

Computational Side

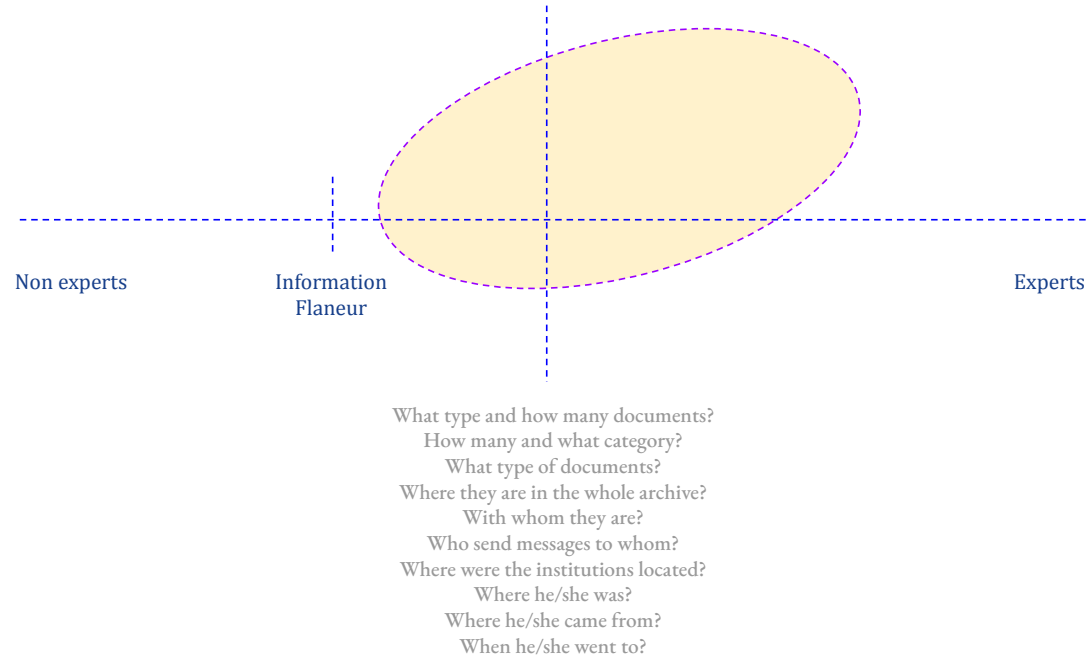


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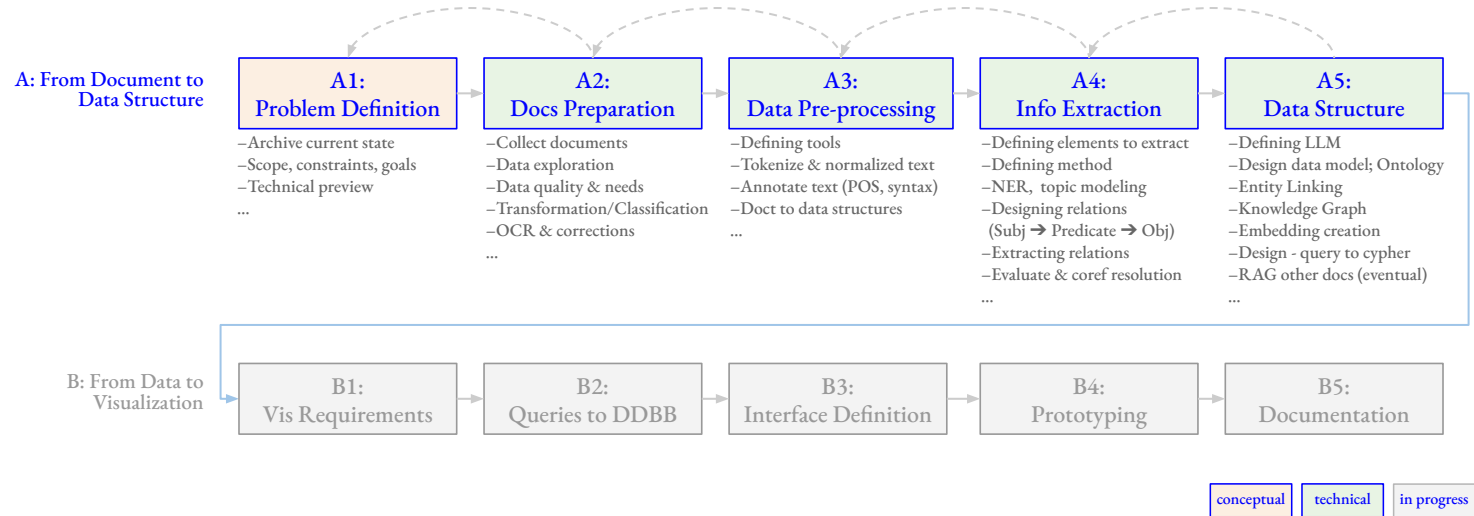
Methodological Side

Mixed Methods: Interviews / Literature / Text Analysis / User Testing ...

Users possible questions



Project Main Tasks & Paths:



35

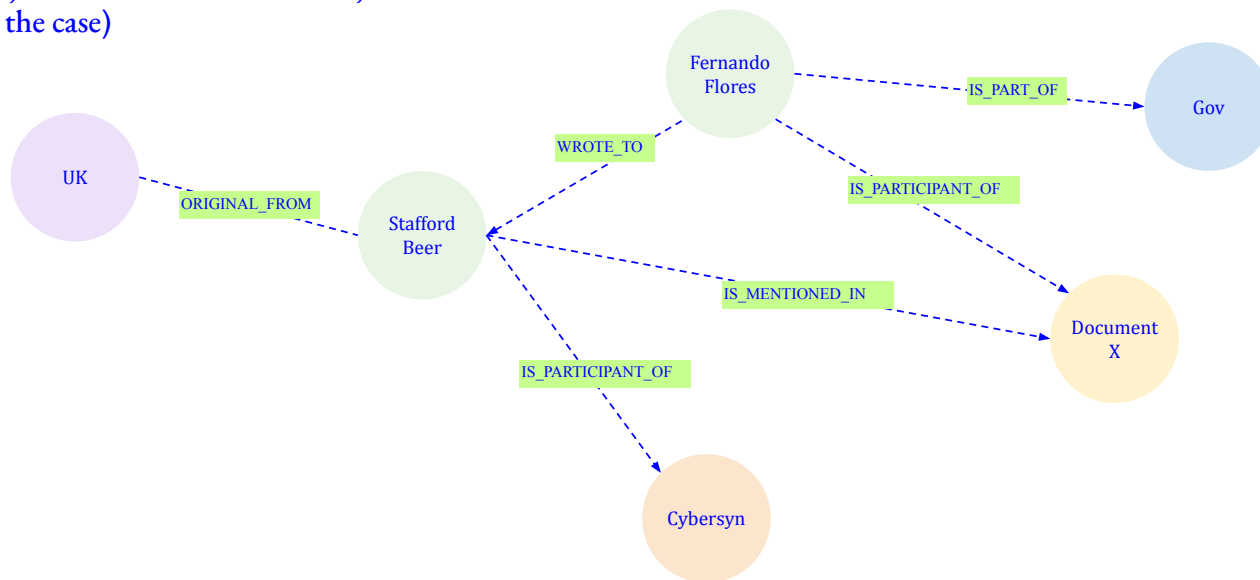
Visualization project
Doing

Auto ethnography & Doc.
Reflecting

Tesis
Writing & Documentation

Why Knowledge Graphs

Triplet “Subject” --> “Action” --> Objects”
(depending the case)



KGs permit representation of complex relationships among different types of entities, such as people, events, places, concepts...

KGs helps to go from a traditional document-centric approach to a more entity-centric approach.

Competency Questions to Create Entities & Relations

Competency Questions

Factual questions (direct properties of an entity):

- Who invited Stafford Beer to Chile?
- Who was in charge of Cyberstryde?
- What components made up Project SYNCO?

Indirect relationships (intermediate connections, cross-referencing documents, organisations, and people):

- Which documents were discussed in the correspondence between Stafford Beer and the Opsroom team?
- Which letters represent Beer's defence of the project against political attacks?

Structural questions (groupings or temporal evolution):

- Which cybernetic concepts appeared most frequently in the documents during 1972?
- Who worked in the OpsRoom?

Hypothetical or causal questions:

- Which individuals linked the government to factory operations within Project Cybersyn?
 - Which document mentions government officials involved in Project Cybersyn?
- INITIAL ONTOLOGIES

Entities & Relations

~~–ENTITIES–~~

PERSON
PLACE
ORGANIZATION
EVENT
PROJECT
CONCEPT
TECHNOLOGY
DOCUMENT

~~–RELATIONS–~~

IS_PARTICIPANT_OF (Actor > Project)
WAS_AT (Actor > Place)
PARTICIPATED_IN (Actor > Project)
WROTE_TO (Actor > Actor)
WORKS_FOR (Actor > Institution)
COLLABORATED_WITH (Actor > Actor)
IS_AUTHOR_OF (Actor > Document)
WAS_MEMBER_OF (Actor > Institution)
IS_RECIPIENT_FROM (Actor > Document)
USED_IN (Technology > Project)
DEVELOPED_BY (Project > Actor)
WAS_USED_IN (Technology > Project)
DEVELOP_CONCEPT (Document > Concept)
SITUATED_IN_PHASE (Document > Operational Phase):
SHARE_TOPIC (Document > Document):
REFERE_TO (Document > Document)
MENTION_TO (Document > Actor)
TREAT_ABOUT (Document > TopicConcept)
IS_PART_OF (Document > CollectionSection)
THIS_LOCATED_IN (Document > Geographic Location):
TRAVELED_TO (Actor > Place)
ORIGINAL_FROM (Actor > Place)
OCCURS_IN (Document > TimePeriodInstant)

Identify main tasks

Distribution

What type and how many documents?
How many and what category?
...

Part-to-a-whole - Proportions

What type of documents?
Where they are in the whole archive?
...

Relationships

With whom they are who send messages to whom else?
...

Location

Where were the institutions located?
Where he/she was?
Where he/she came from?
When he/she went to?
...

Main Challenges

- Relationships between documents/actors:

User can follow relations among documents, actors, dates, etc.
(Schich et al. 2014; Aiello y Simeone, 2019),

- Summaries and Metadata:

User will get some information from the database
(Kavaz et al., 2023)

-Interactions and Search:

Querying/searching from prompt box to the database and documents (Zhang et al., 2023; Dabek and Caban, 2017).

-Chat to Viz:

Respond to user requests to generate graphs.
(Kavaz et al., 2023).

-Generative Visualizations:

Generating a visualization (Shen et al., 2021).

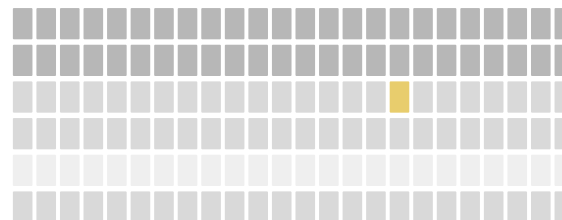
Visualization levels & scopes

Distribution

What type and how many documents?
How many and what category?
...



Macro - whole archive
structure, categories,
clusters



Part-to-a-whole - Proportions

What type of documents?
Where they are in the whole archive?
...



Relationships

With whom they are
who send messages to whom else?
...



Meso - categories/clusters
documents, persons,
institutions
relations, positions, etc.



Location

Where were the institutions
located?
Where he/she was?
Where he/she came from?
When he/she went to?
...

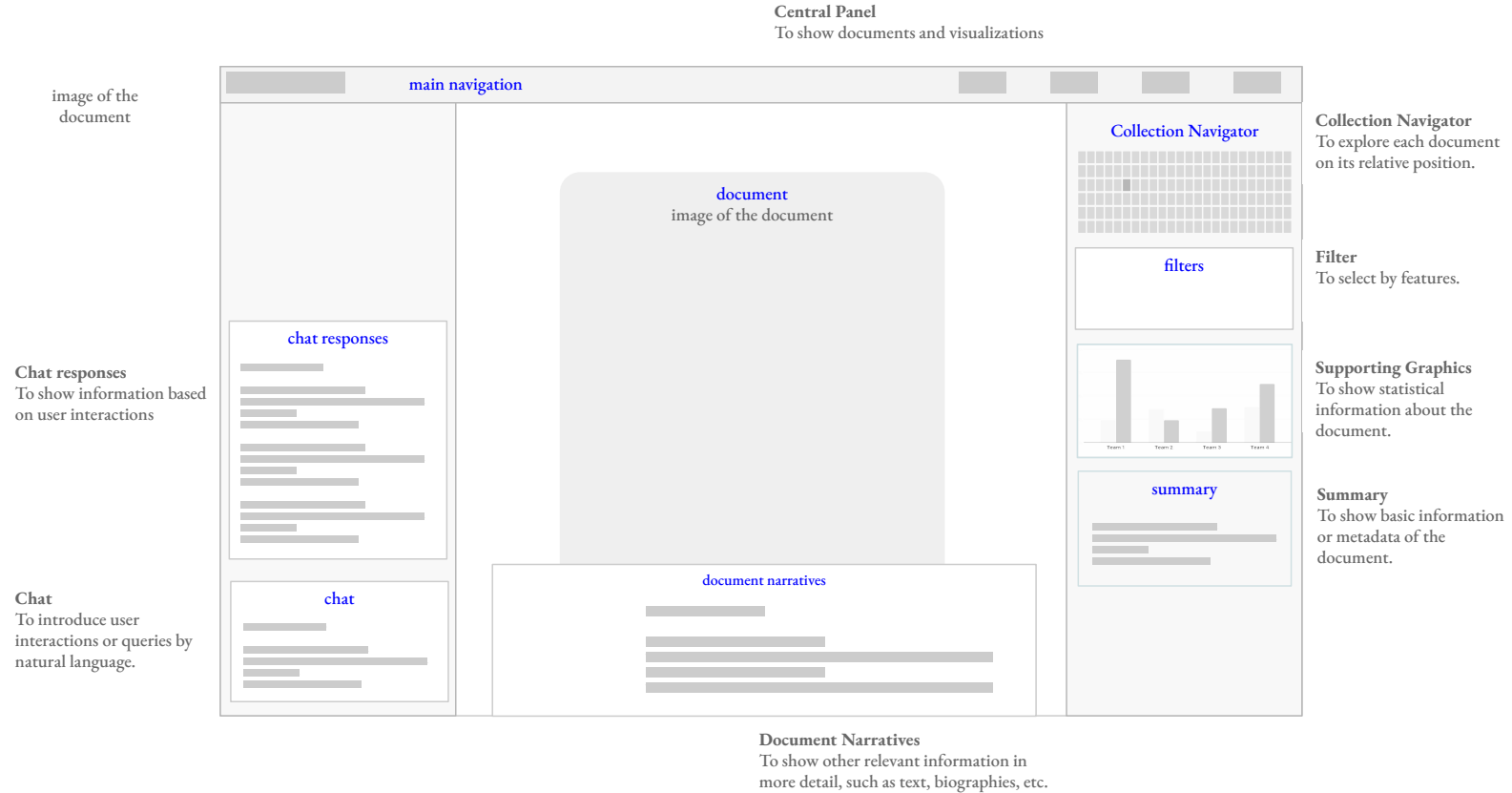


Micro - single entity...
documents, persons,
institutions relations,
positions, etc.



Project Proposal

Possible Layout for elements



Project Status

Prototyping text extraction from image (Tesseract)



3rd August 1973

Dear Raul

It has taken some time for me to deliberate the enclosed letter to the President.

I have said in it only those things which my professional conscience dictates must be said.

Therefore I am relying on you to explain the necessity of the letter to Fernando, since after all a statement was demanded of me by the President ; and then to transmit the letter with an appropriate translation.

Yours ever

Prototyping image description (GPT, Gemini)



Esta es una fotografía histórica en blanco y negro que captura a un grupo de figuras clave relacionadas con el Proyecto Cybersyn en Chile.

A continuación, describo los elementos principales de la imagen:

Sujetos y Contexto Histórico

Personajes: De izquierda a derecha, se observa a Gui Bonsiepe (diseñador industrial alemán), seguido por Stafford Beer (el cibernético británico, fumando), Herman Schwember y Fernando Flores (ministro de economía y pieza central del proyecto).

Ubicación: El pie de foto indica que la escena tiene lugar en el Restaurante Cura Nurin, en Viña del Mar (Chile).

Fecha: La imagen está datada en 1971, durante el gobierno de Salvador Allende.

Composición y Estilo

Acción: Los cuatro hombres aparecen caminando al aire libre en lo que parece ser una conversación informal. La actitud es relajada; Stafford Beer sostiene lo que parece ser un cigarrillo o puro cerca de su boca.

Vestimenta: Refleja la época y un estilo intelectual/informal. Se observan cardiganes de punto, chaquetas ligeras, camisas abotonadas y anteojos de pasta gruesa.

Entorno: Al fondo se aprecia un paisaje con vegetación, una estructura de piedra (posiblemente un muro o entrada) y una casa de estilo clásico sobre una colina. También se ve un vehículo antiguo estacionado a la izquierda.

Prototyping entity extraction (GPT, Gemini).



```
img_text = """
3rd August 1873
```

Dear Raúl

It has taken some time for me to deliberate the enclosed letter to the President.

I have said in it only those things which my professional conscience dictates must be said.

Therefore I am relying on you to explain the necessity of the letter to Fernando, since after all a statement was demanded of me by the President ; and then to transmit the letter with an appropriate translation.

Yours ever
"""

```
prompt_img = """From the text provided between accents\
contains or deals with a photo description, you will identify
the following:
```

1. Identify if it is a photo description (true boolean - is_description)
2. Identify the people mentioned (people)
3. Identify the places mentioned (places)
4. Identify the organizations mentioned (organization)
5. Identify dates (dates)
6. Extract 3 keywords, 1 word per keyword (keywords)

```
If it does not contain people, write "no people"
If it does not contain places, write "no places"
If it does not contain dates, write "no dates"
If it does not contain organization, write "no organization"
```

```
Deliver the response as a JSON object with the following names:
is_description:, people:, places:, dates:, organization:,
Keywords:,
```

```
If the provided text does NOT contain or DOES NOT It's about an
photo, simply write "It's not about photo."
```

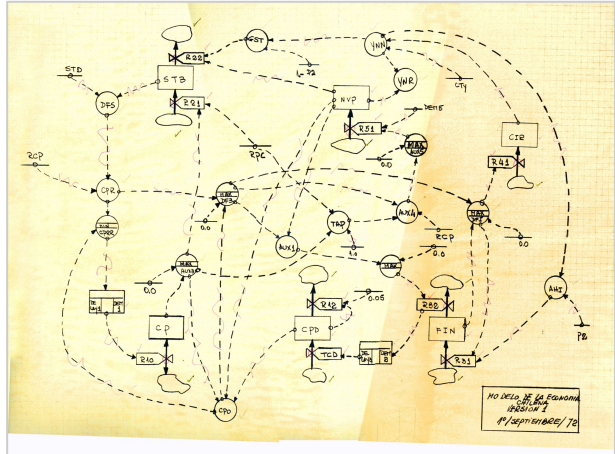
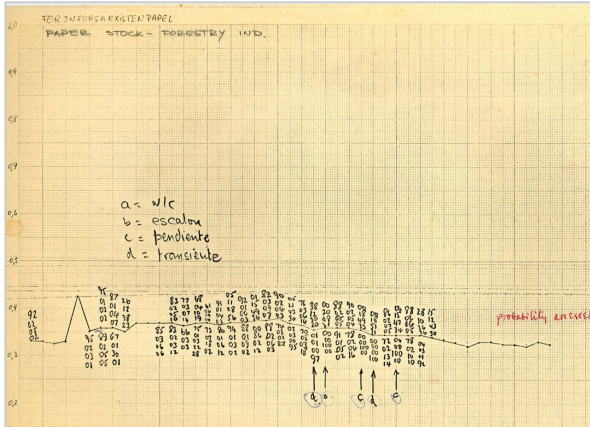
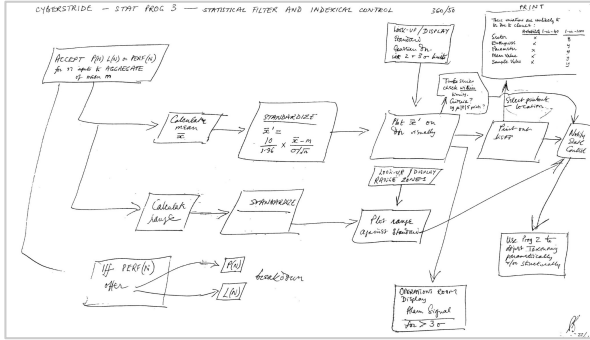
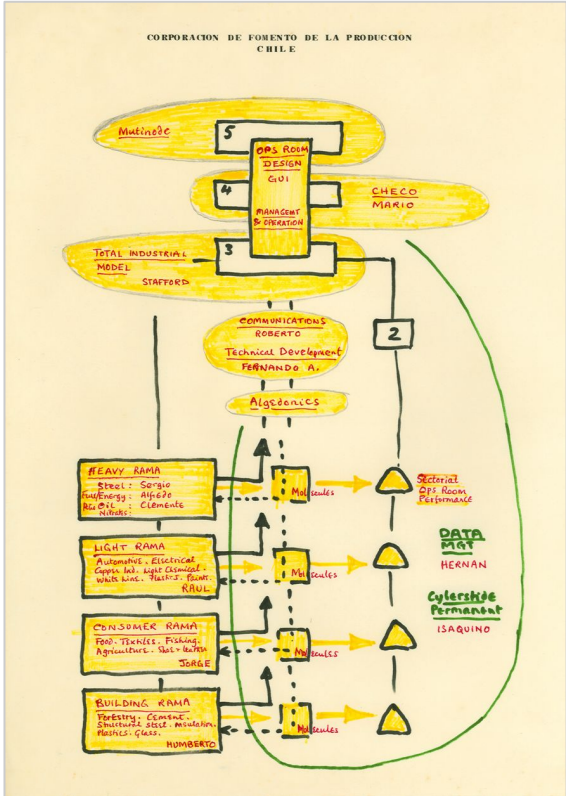
```
```{text_to_json}```
"""
```

```
resp_prompt_ent1 = obtener_respuesta(prompt5_ent1.format(
mi_textito = img_text))
```

```
print(resp_prompt_ent1)
```

```
Título: No hay título
Personas: Raúl, Fernando
Lugares: No hay lugares
Fechas: 3rd August 1873
```

## Challenges



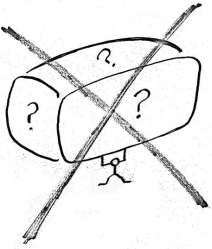
# Challenges

3 First Principle : GOVERNMENT IS THE PEOPLE'S HELP

FINISHED

The view of government

- as incomprehensible
- as the people's burden



STOP THIS ↑ WHENEVER YOU CAN

Nunca desistir

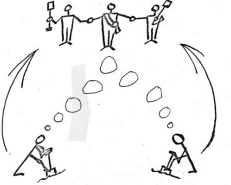
CHANGE is a state of mind that everyone shares

4

STARTED

The view of government

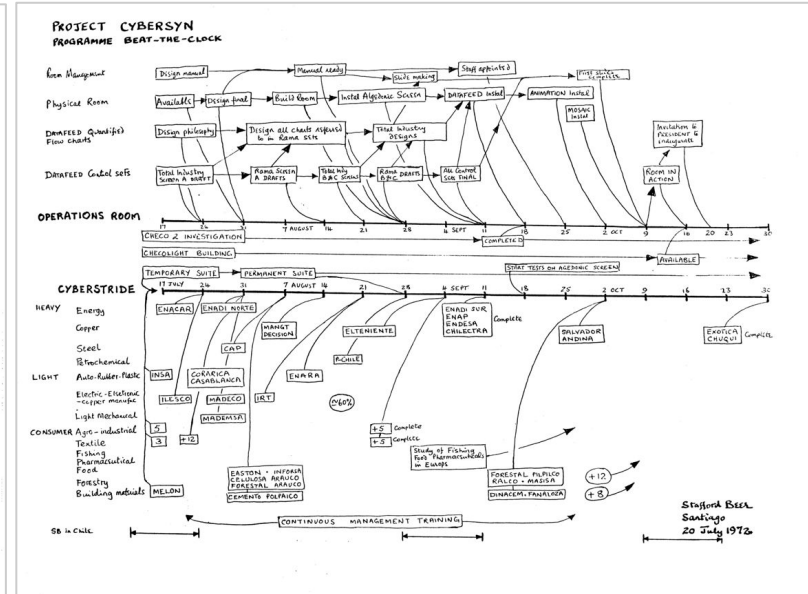
- as the people thinking what to do
- as acting together to do it



The Goparito President says : "Government is the people".

The crises of the people will be made known to the Government at all times.

We share the TECHNOLOGY, which belongs to the people, to do it.



## Challenges

*Proposed program for developing Dyn. Mod. of Economy*  
 PROGRAMA PROPUESTO PARA DESARROLLAR EL MODELO DINAMICO DE LA ECONOMIA.

### 1. Estudio exhaustivo del Checo 1.

Implicaría: *study in detail*

1.1 Estudiar a fondo la Teoría de la Dinámica Industrial.

1.2 Estudiar a fondo el Modelo Económico planteado.

1.3 Practicar con el CHECO 1, con el fin de adquirir familiaridad en el manejo e interpretación de las Variables de Proceso, Parámetros y Resultados, así como una comprensión de la Dinámica del Proceso en General.

1.4 Determinar con mayor precisión las Constantes, Variables y parámetros que resulten de mayor relevancia en el Modelo.

1.5 Preparación de un Equipo destinado a abordar el Problema Computacional involucrado.

### 2. Modificaciones al Checo 1.

Una vez adquirido el dominio del CHECO 1, la etapa siguiente consistiría en evolucionarlo hasta un Modelo que represente mejor nuestra Economía. Implicaría:

2.1 Replanteamiento del CHECO 1 a la luz de la sensibilidad de los Factores e Interacciones más significativos.

2.2 Análisis de aquellas Áreas del Modelo que requieren de una mayor precisión. Considerando la posibilidad de desarrollar Modelos Complementarios para dichas Áreas.

2.3 Experimentación con el Modelo Modificado.

2.4 Pronunciamiento del Equipo de Computación sobre la Naturaleza del Compilador requerido y de la Problemática asociada.

3. Transferencia de Software.

3.1 Adquirir el Compilador adecuado para la utilización efectiva de la Dinámica Industrial.

*Building Compiler*

CORPORACION DE FOMENTO DE LA PRODUCCION  
CHILE

### THE ROOM

CORFO to provide plans of the room in all nine dimensions.

The Operations Room should be thought of

*NOT* as a room containing interesting bits of equipment

*BUT* as a control machine comprising men and artifacts in symbiotic relationship

It needs designing as a totality and as an operational entity.

This is a problem in ergonomics.

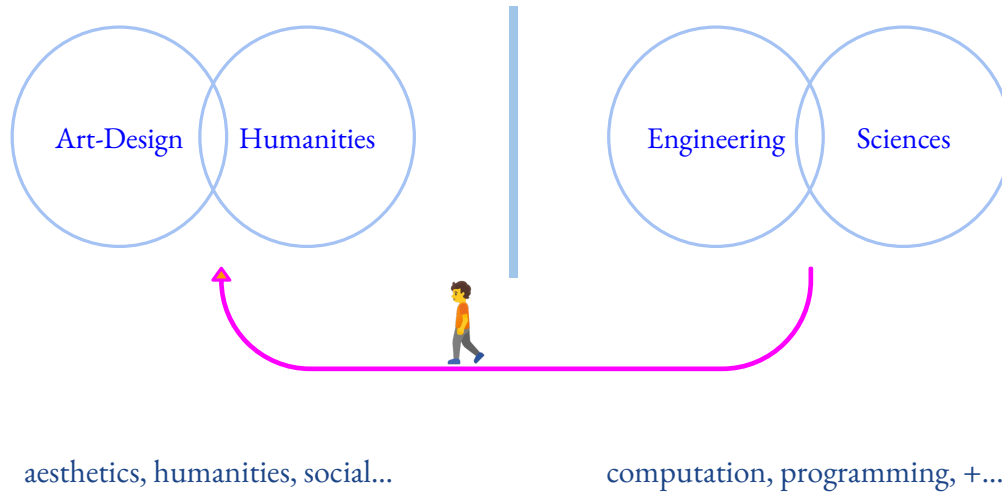
A three-dimensional model of the room will be gradually constructed in London with the help of Gui Bonsieppe in Chile.

We will keep in touch by Telsax and especially by exchanging photographs.

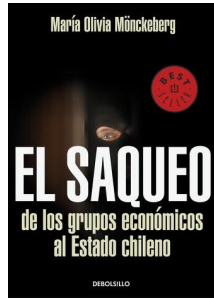
Special attention will be paid to the development of man-machine interfaces, using 'big hand' controls.

# *Personal Motivation*

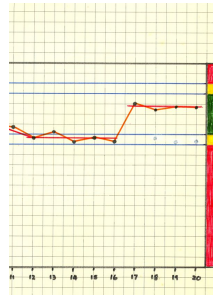
## What to do with this knowledge & experience



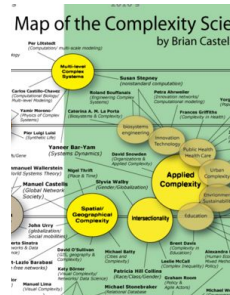
## Motivation / Projections



**Book Visualization**  
"El saqueo de los grupos económicos al estado de Chile", 2015. María Olivia Mönckeberg.



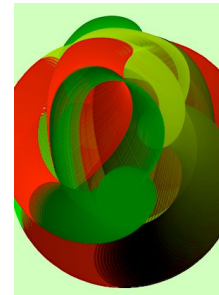
**Cultural Documents**  
Reconstructing the diagrams of the Cybersyn project



**Scientific Information**  
Castellani, B. (2019). Map of the complexity sciences. [art-sciencefactory.com/complexity-map\\_feb09.html](http://art-sciencefactory.com/complexity-map_feb09.html)



**Forensic Visualization**  
DDHH related issues.



**Computational art**  
Personal

**Propuesta Constitución 2022**  
*Consolidado normas aprobadas para la pleno de la Convención*

Este documento es un borrador por consultar. La redacción definitiva, ya que es pre-proyecto.

CAPÍTULO PRIMERO		CAPÍTULO SEGUNDO		CAPÍTULO TERCERO		CAPÍTULO CUARTO		CAPÍTULO QUINTO	
ARTÍCULO	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1
8	1	1	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1	1	1
10	1	1	1	1	1	1	1	1	1
11	1	1	1	1	1	1	1	1	1
12	1	1	1	1	1	1	1	1	1
13	1	1	1	1	1	1	1	1	1
14	1	1	1	1	1	1	1	1	1
15	1	1	1	1	1	1	1	1	1
16	1	1	1	1	1	1	1	1	1
17	1	1	1	1	1	1	1	1	1
18	1	1	1	1	1	1	1	1	1
19	1	1	1	1	1	1	1	1	1
20	1	1	1	1	1	1	1	1	1
21	1	1	1	1	1	1	1	1	1

**Public Information**  
Adding more constitutions:  
Chile 1925, 1980, 2022.  
France, USA, Germany (1948)  
etc...

## Motivation

# Visualization is a tool to challenge *Information asymmetries*

*in*

public information  
scientific information  
cultural information  
humanities

...

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Thank You

Comments & Questions

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From Document to Meaningful Exploration (in progress):

# *Visualising Cybersyn Documents*

Ricardo Vega Mora  
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University of Edinburgh (Professor: Melissa Terras & John Vines)

SB100Y Conference  
18, Sept, 2026

NEW FOR UTA, JOHN, MELISSA

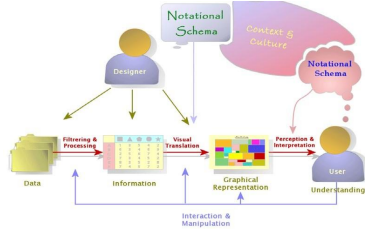


Ricardo Vega Mora  
PhD Student in Computer Science Engineering  
Pontificia Universidad Católica de Chile.  
Professor: Juan Reutter

## Possible outputs

	question	methods	user	output
a) Analysis of visualization models and/or books	How have LLMs and Generative AI changed the tasks of the visualization process?	—+30 models & +25 books —model analysis & comparison —main task identified	Academics and practitioners who need to understand the visualization process.	Account of differences, challenges and opportunities for visualization process using AIGen/LLMs
b) Visualization prototyping using AI/LLMs methods	How can LLM and GenAI be implemented to automate and simplify the visualization process and thus facilitate document exploration?	—documents processing —identify main tasks —tasks prototyping	Academics and practitioners who need to understand the visualization process.	A case study visualisation prototype with implementation assisted by AI/LLMs
c) Documenting and reflecting the process.	What are the methods to register and document the use of LLM and AI in the process of visualization?	—Autoethnography tools —Drawing, sketching —visualization sketching	Academics and practitioners who need to understand the visualization process.	Online - offline documentation open to the public.

## Auto ethnography: Registering tasks (initial proposal)



**Task:** Apply Named Entity Recognition

**Phase:** A: From Document to Data Structure.: Apply Named Entity Recognition

**Description:** For a given text, Its goal is to recognize the entities ...

**Start:** as input, it requires any text file containing pure characters.

**Result:** As output a file or structured with the required entities extracted.

**Observations:** frequently the result required iterations and a process of cleaning...

**Classic Methods:** usually from handwriting observation and extraction.

**AI Methods:**

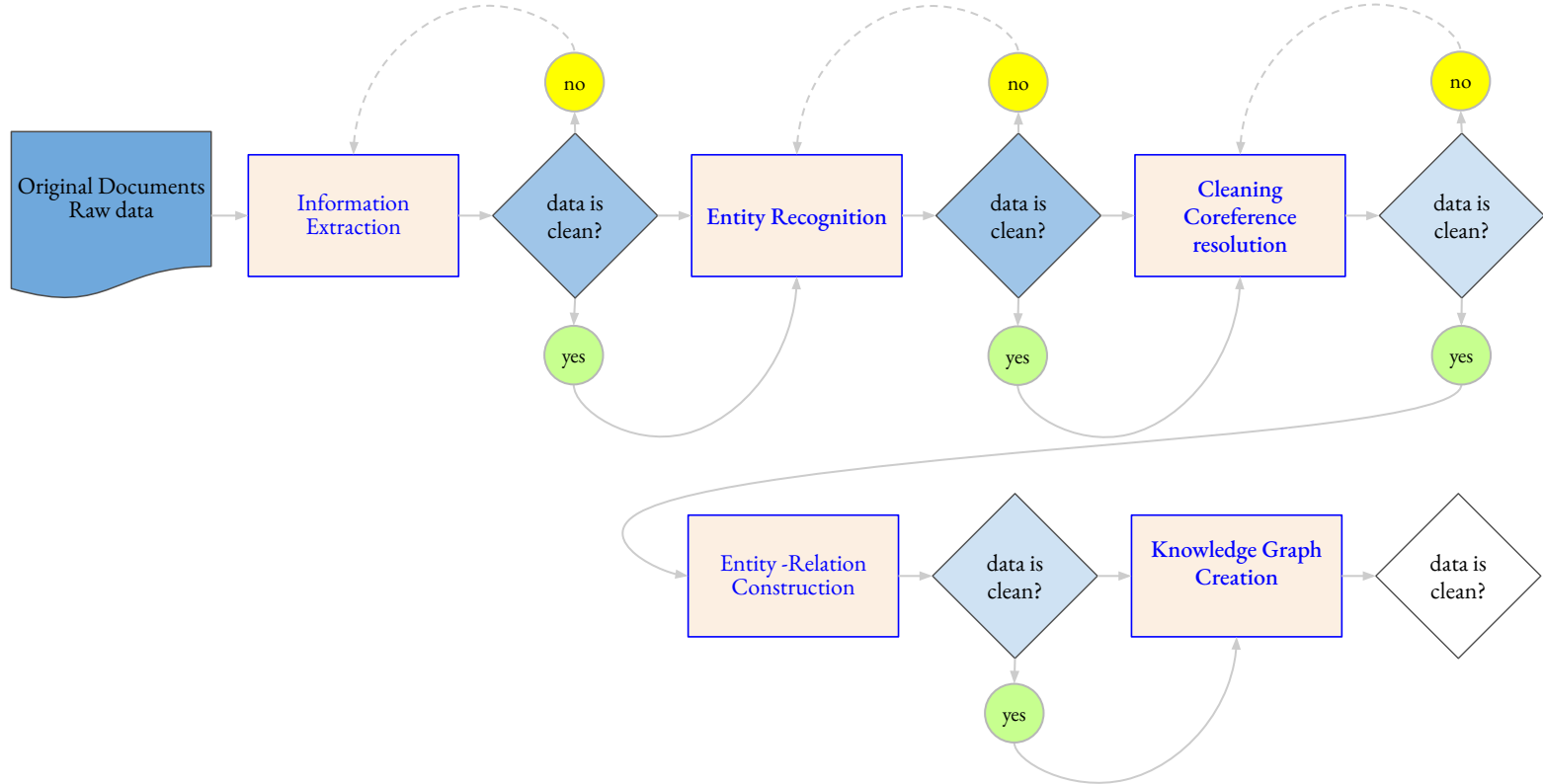
**Differences/Similarities:**

**Experience:** it was frustrating sometimes using traditional method...

**Recommendations:** sometimes a mixture among using LLMs, Gen AI & classic...

→ *Analysis*

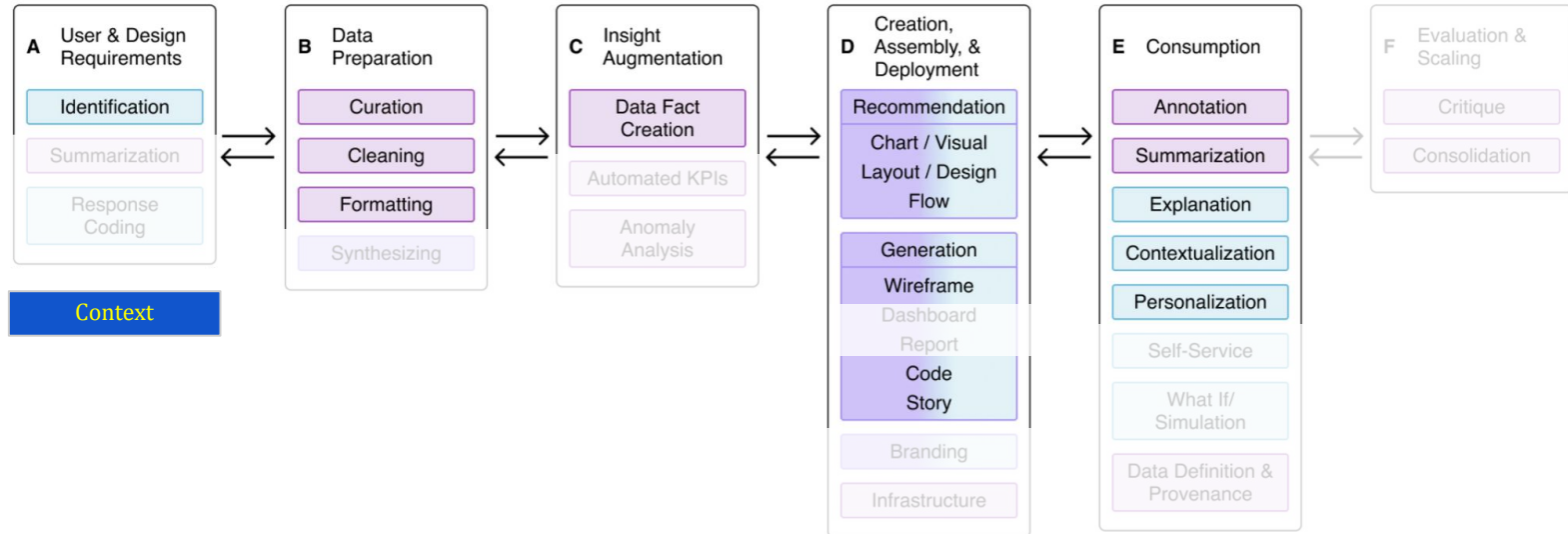
## from document to KG:



## Diagram and Tasks: Priorities

### Visualization Workflow Phases

Types of Generative AI: ■ Creativity ■ Assistance ■ Automation



## VisHub Presentation:

Question: How has visualization changed since the emergence of Artificial Intelligence?

Where I come from... (ternsionstechnique-representation)

C80: Using visual resources

Questions I had / Needs observed

I found a paper...

Big frame of the proposal

Proposal, questions

Relevance & Research Contribution

Visualization Main Roles

Visualization The broader visualization spectrum...

Visualization Standard graphics vs custom made:

Visualization Models: How to understand visualization tasks

State of the Art: AI and Visualization current research - Two sides: VIs4AI - AI4Vis

What is cybernetics

Case Study: Stafford Beer Archive, Liverpool John Moores University (LJMU)

Case Study: SBA (LJMU) - Structure

Case Study: SBA (LJMU) - Available Boxes

Case Study: SBA (LJMU) - File Types

Methodology

Methodology pipeline

How Visualization is produced

Possible outputs

Auto ethnography: Registering tasks (initial proposal)

Pipeline, Tasks:

Help to understand what im doing

## Plan/Gantt Chart



		2025-1 Chile			2025-2 Chile			2026-1 Abroad			2026-2 Abroad			2027-1 Chile		
Questions	Mayor Tasks	Jan Feb	Mar Apr	May Jun	Jul Agu	Sep Oct	Nov Dec	Jan Feb	Mar Apr	May Jun	Jul Agu	Sep Oct	Nov Dec	Jan Feb	Mar Apr	May Jun
Stage A: Chile From Data to Analysis . P01, P02	A1- Visualization Models research	x	x	x	x											
	A2- Data Preprocessing			x	x	x	x									
	A3-Prototyping Tasks			x	x	x	x									
	A4- Cultural Analytics							x	x	x						
	A5- Knowledge graph DDBB v1							x	x	x						
	A6- Results/Evaluation								x							
Stage B: Edinburgh, Scotland. From Data to Vis. Back in Chile. P1, P03	B1- Knowledge graph DDBB v2								x	x	x	x				
	B2- Visualization development									x	x	x	x			
	B3- Functional Prototype									x	x	x	x	x	x	
	B4- Results/Evaluation														x	x
	Tesis										x	x	x	x	x	x



(ANID )

Accepted for Double Degree

Informatics: ILCC: Language Processing, Speech  
Technology, Information Retrieval, Cognition  
PhD, College of Science & Engineering  
School of Informatics



**John Vines (Main tutor):**

Co-director Institute for Design Informatics, Director of  
the Centre for Design Responsible Natural Language  
Processing. [www.designinformatics.org/person/john-vines](http://www.designinformatics.org/person/john-vines)



**Melissa Terras (Co tutor):**

Director of Center for Data, Culture & Society of  
Edinburgh  
<https://www.cdcs.ed.ac.uk>

## Limitations

- The approach to the process is general
- Do not delve into any specific technical or methodological aspect.
- The project explores the process empirically
- For desktop web platform
- Simple usability and user experience evaluations are considered
- Limited to specific tasks
- Time...
- ...

## Potential concerns or objections

- Limited testing
- Lack of specific focus
- Need more case testing
- Focused on specific tasks
- ...

## Proposal

### Research Question

How can Large Language Models and Generative Artificial Intelligence be implemented in the visualization process to automate and simplify work with multimodal documents, enabling greater responsiveness to user interactions in the humanities context?

### Hypothesis:

Large Language Models and Generative Artificial Intelligence can be implemented to automate the visualization process of multimodal documents in the humanities, enabling data processing to be simplified and, in turn, enhancing responsiveness to user interactions through the on-demand generation of contextual and dynamic visualizations.

### Main goal

To identify the possibilities and opportunities that Large Language Models and Generative Artificial Intelligence offer to the data visualization process in the humanities context by prototyping a multimodal document visualization project.

## Proposal

### Research Question

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### Questions

**Q01:** How can we generate a conceptual model for processing and visualizing cultural documents that incorporates Big LLMs and Generative AI, facilitating the automation of processes within the humanities?

**Q02:** How can we design multidisciplinary strategies for analyzing cultural documents using computational methods such as Big LLMs and Generative AI within the humanities?

**Q03:** How can we enable the interactive and meaningful exploration of multimodal documents through visualization assisted by Big LLMs and Generative AI?

### Hypothesis

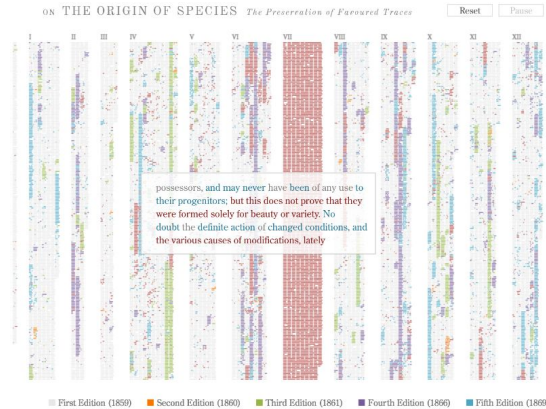
**H1:** The most recognized conceptual models describing the data visualization process can expand their descriptive capacity by incorporating LLM and Generative AI into their descriptions.

**H2:** Developments in LLM and Generative AI accelerate and streamline the various stages of the multimodal document visualization process.

**H3:** The use of Language Models and Generative AI facilitates the exploration of multimodal documents with on-demand generative visualizations.

## Visualization

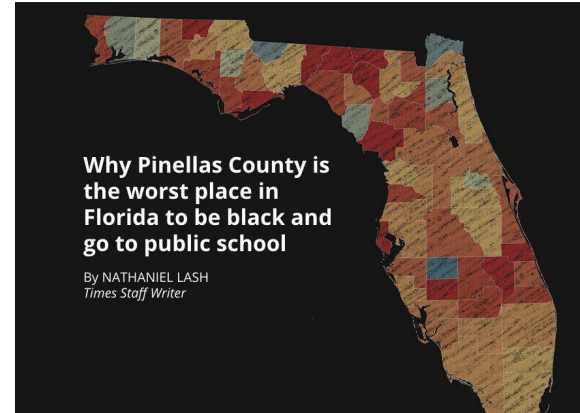
### Custom made examples



#### Traces, Ben Fry (2010)

Exploration of the editions of Charles Darwin's "On the Origin of Species." A visual exploration not found in standardized bookstores.

link: <https://fathom.info/traces/>



#### Why Pinellas County is the worst place in Florida to be black and go to public school

It integrates graphics to tell a complex story.

link:

[projects.tampabay.com/projects/2015/investigations/pinellas-failure-factories/chart-failing-black-students/#](https://projects.tampabay.com/projects/2015/investigations/pinellas-failure-factories/chart-failing-black-students/#)

I found a paper...

“Recent developments in artificial intelligence (AI) and machine learning (ML) have led to the creation of powerful generative AI methods and tools capable of producing text, code, images, and other media in response to user prompts.”

66

“However, there remains a **lack of understanding** about **which visualization activities** may be particularly suitable for the application of generative AI”

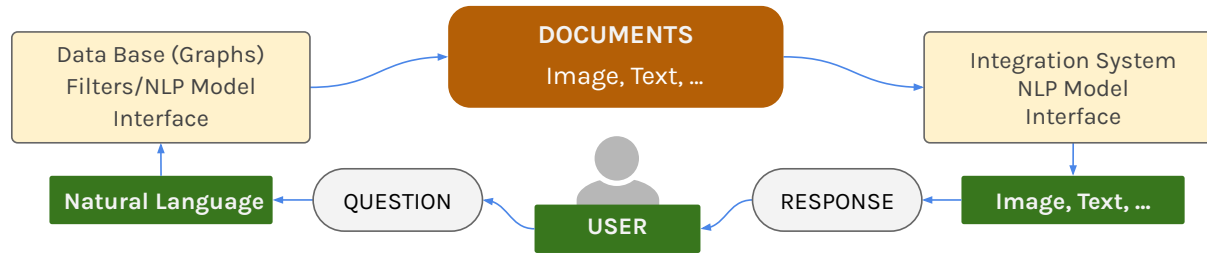
How to use AI in data processing  
and visualization?



## Case Study: SBA (LJMU) Technological, Cultural, & Computational Interests.

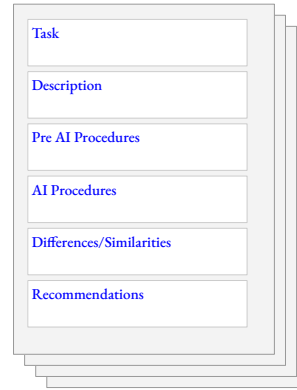
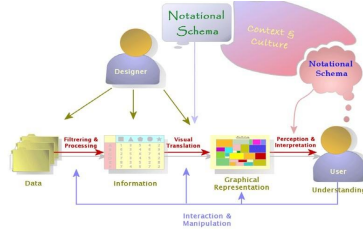


## Methodology



What type and how many documents?  
 How many and what category?  
 What type of documents?  
 Where they are in the whole archive?  
 With whom they are?  
 Who send messages to whom?  
 Where were the institutions located?  
 Where he/she was?  
 Where he/she came from?  
 When he/she went to?

## Registering tasks (initial proposal)



*Analysis*

## Visualization options

### What to show?

#### Distribution

What type and how many documents?

How many and what category?  
...

#### Part-to-a-whole - Proportions

What type of documents?

Where they are in the whole archive?  
...

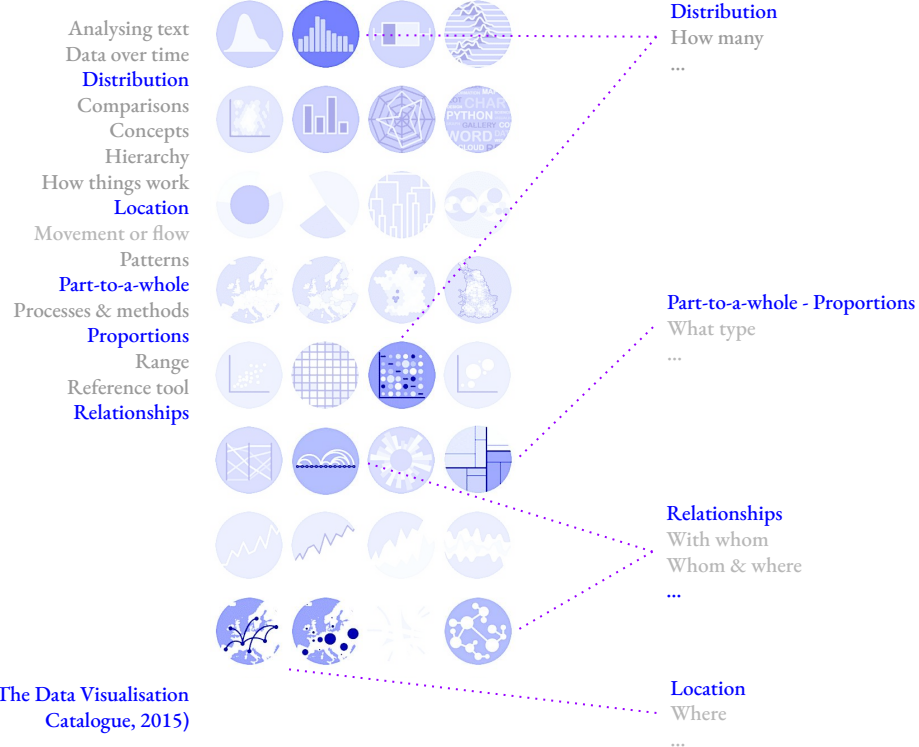
#### Relationships

With whom they are who send messages to whom else?  
...

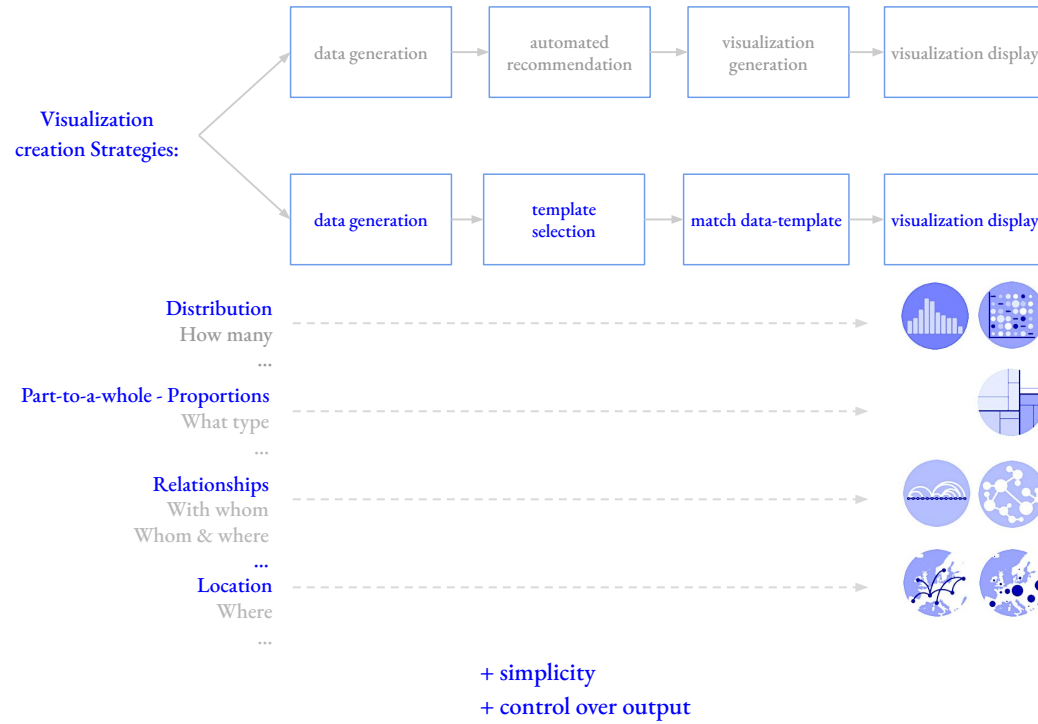
#### Location

Where were the institutions located?

Where he/she was?  
Where he/she came from?  
When he/she went to?  
...

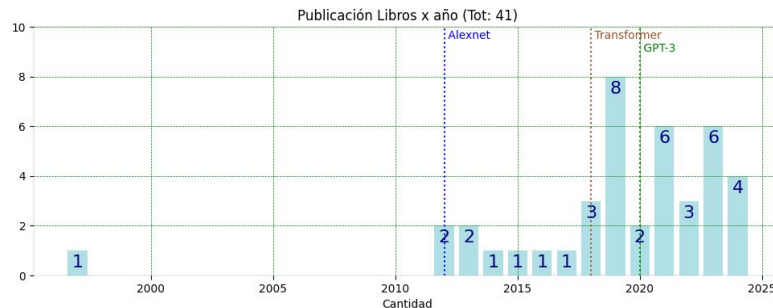


## How Visualization is produced



## Visualization books and authors

### To understand how visualization has changed



#### Description:

41 Data visualization books - from 1997 - 2024

Searched in web repositories & Repository UC by keywords: data visualization, Information visualization, infovis, dataviz, data representation, etc.

#### Initial conclusions:

Great interest from 2017, but no mention to AI/LLM in the most recent books,

The impact of AI on Vis remains within the academic sphere yet.

1997 The Visual Display of Quantitative Information  
 2012 The Functional Art: An introduction to information graphics and visualization (Voices that matter)  
 2012 Information Is Beautiful  
 2013 Interactive Data Visualization for the Web: An Introduction to Designing with D3  
 2013 Cool Infographics: Effective Communication with Data Visualization and Design  
 2014 Visualization Analysis and Design (AK Peters Visualization Series)  
 2015 Storytelling with Data: A Data Visualization Guide for Business Professionals  
 2016 Dear Data: A Friendship in 52 Weeks of Postcards  
 2017 The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios  
 2018 Data Visualization: A Practical Introduction  
 2018 Making Data Visual: A Practical Guide to Using Visualization for Insight  
 2018 Data Visualization Made Simple  
 2019 Fundamentals of Data Visualization: A Primer on Making Informative and Compelling  
 2019 Data Visualization: A Handbook for Data Driven Design  
 2019 Fundamentals of Data Visualization: A primer on making informative and compelling figures  
 2019 Storytelling with Data: Let's Practice!  
 2019 Effective Data Visualization: The Right Chart for the Right Data  
 2019 Info We Trust: How to Inspire the World with Data  
 2019 How Charts Lie: Getting Smarter about Visual Information  
 2019 Effective Data Storytelling: How to Drive Change with Data, Narrative and Visuals  
 2020 Information Visualization: Perception for Design (Interactive Technologies)  
 2020 Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python  
 2021 Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks  
 2021 Data Sketches: A Journey of Imagination, exploration, and beautiful data visualizations  
 2021 Hands-On Data Visualization: Interactive Storytelling From Spreadsheets to Code  
 2021 Data Visualization in Python with Pandas and Matplotlib  
 2021 Data Visualization with Excel Dashboards and Reports  
 2021 Now You See It: An Introduction to Visual Data Sensemaking  
 2022 Functional Aesthetics for Data Visualization  
 2022 Data Engineering with Alteryx: Helping data engineers apply DataOps practices with Alteryx  
 2022 Persuading with Data: A Guide to Designing, Delivering, and Defending Your Data  
 2023 Data Visualization with Python and JavaScript: Scrape, Clean, Explore, and Transform Your Data  
 2023 R for Data Science: Import, Tidy, Transform, Visualize, and Model Data  
 2023 The Art of Insight: How Great Visualization Designers Think  
 2023 Information Design Unbound: Key Concepts and Skills for Making Sense in a Changing World  
 2023 Data Visualization with Python: Exploring Matplotlib, Seaborn, and Bokeh for Interactive Visualizations  
 2023 Python Data Science Handbook: Essential Tools for Working with Data  
 2024 Visualize This: The FlowingData Guide to Design, Visualization, and Statistics  
 2024 Data Visualization with Microsoft Power BI: How to design Savvy Dashboards  
 2024 Data Architect: Building Scalable and Intelligent Data Systems  
 2024 Ultimate Pandas for Data Manipulation and Visualization: Efficiently Process and Visualize Data with Python's Most Popular Data Manipulation Library

## To understand visualization as multidisciplinary field (in progress)



Colin Ware  
Alberto Cairo  
Alex Kolokolov, Maxim Zelensky  
Andy Kirk  
Brent Dykes  
Claus O. Wilke  
Cole Nussbaumer Knaflic  
Daniel Fisher, Miriah Meyer  
David Landup  
David McCandless  
Dick Kusleika  
Dr Pooya  
Edward R. Tufte  
Giorgia Luppi, Stefanie Posavec  
Hadley Wickham  
Jack Dougherty, Ilya Ilyankov  
Jake VanderPlas  
Jonathan Schabish  
Kieran Healy  
Kristen Soudki  
Kyrán Dale  
Miro Kazarokoff  
Nadia Breffner, Shirley Wu  
Nathan Yau  
Paul Houghton  
Peter Bruce, Peter Gedeck  
Randy Krum  
RJ Andrews  
Sam Green  
Scott Murray  
Sheila Pontis, Michael Babwahsin  
Stephanie Evergreen  
Stephen Few  
Steve Wexler  
Tahera Firdose  
Tamara Munzner  
Vidya Setlur

ciencia computacional, ingeniería marina  
diseño, periodismo  
educación, inteligencia empresarial  
diseño, consultoría, autor  
análisis de datos, comunicación  
biología computacional, biología evolutiva  
matemáticas aplicadas  
diseño, consultoría, ciencia computacional, astronomía-astrofísica  
ingeniería de software, robótica  
periodismo, diseño  
contaduría  
ingeniería, ciencia computacional  
estadística, arte, educación  
arte, autor  
estadística, biología humana  
filosofía, estudios sociales, estudios políticos, ciencia computacional, artes plásticas  
astronomía-astrofísica  
economía, escritura, educación  
sociología  
análisis empresarial, programación, visualización  
filosofía, robótica, neurociencia computacional  
ciencia computacional, comunicación  
astronomía-astrofísica, ciencia de datos, ingeniería de software, diseño  
estadística  
análisis empresarial  
ciencia de datos, ciencia computacional, química  
diseño  
administración de empresas, ingeniería en ciencias mecánicas  
ingeniería, diseño, educación  
diseño  
diseño, investigación  
diseño, investigación  
consultoría, inteligencia empresarial  
ciencia en informática, consultoría, autor  
educación matemática  
ciencia computacional, investigación, visualización  
Investigación, estadística, gráficas computacionales

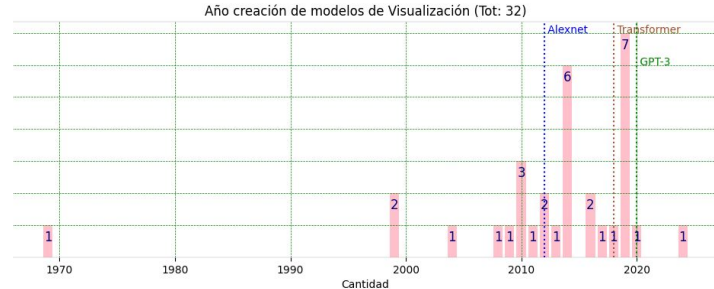
44 unique Data visualization books authors.

Many authors start in one discipline and larger mode to other, for example:

psychology (undergrad) → comp. sciences (master)  
comp. sciences (undergrad) → public policy (master)

This can show the multidisciplinary nature of visualization.

## Visualization Conceptual Models: To understand stages and common tasks (in progress)



### Description:

32 Data visualization Models - from 1970 - 2024

Searched in books, thesis & papers.

### Initial conclusions:

Discipline of authors is relevant for the main focus of the model (computational, narrative, etc). Mainly describing the visualization process de  
Until 2024, no mention to AI/LLM.

The models do not integrate AI on Visualization yet.

(just a recent one, Basole et al in 2024)

### visualization conceptual models

Visual Data Exploration - Visual Analytics Process Model  
A Nested Model for Visualization Design and Validation  
Knowledge Generation Model for Visual Analytics  
Data Visualization Construction and Interpretation  
The Hermeneutic Circle of Data Visualization  
Info We Trust  
The Four Stages of the Visualization Workflow  
Seven-step process  
Workflow, Moritz Stefaner  
Information Visualization Framework  
Data Vis Workflow  
Data, head, hand, drawing  
From Data to Knowledge, Density Design Research Group  
The Keys to a Successful Data Design Process  
Five schemas from the book  
InfoVis Diagram  
Design is a Search Problem  
Constructing Charts and Graphs  
Nine-stage design study methodology framework  
Nine-stage design study methodology framework  
Three-part analysis framework for a vis instance  
Nine-stage design study methodology framework  
The Functional Art  
The process of graphic journalism for breaking news  
InfoVis Diagram  
Practical Charting Techniques  
The Hermeneutic Circle of Data Visualization  
Data Visualization Construction and Interpretation  
Data Visualization Construction and Interpretation  
Visualization Process  
Knowledge Crystallization (a)  
Knowledge Crystallization (b)  
Dynamic Visualization Pipeline  
Knowledge Crystallization  
modelo sin nombre  
modelo sin nombre  
modelo sin nombre  
Design process  
The Data Design Guide  
Visual Thinking  
A three-stage model of human visual information processing  
Design Process, EMAPS project

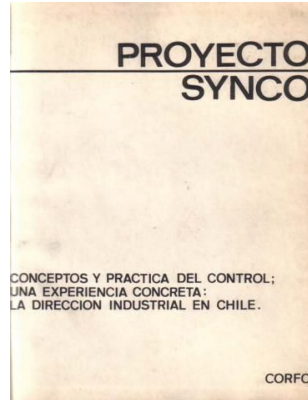
### model authors

Keim, D.  
Munzner, T.  
Sachs, D.  
Ginda, M.  
Rodighiero, D. & Romele, A.  
Andrews, R.J.  
Kirk, A.  
Fry, B.  
Bilhanic, D.  
Lima, M.  
Rost, L.  
Lupi, G.  
Masud, L.  
Murray, S.  
no author  
Dürsteler, J.C.  
Bostock, M.  
Heer, J.  
Meyer, M.  
Munzner, T.  
Munzner, T.  
Sedlmair, M.  
Cairo, A.  
de la Torre Arenas, I.  
Engelhardt, Y.  
Spear, M.E.  
Romele, A.  
Börner, K.  
Bueckle, A.  
Yau, N.  
Card, S.K.  
Mackinlay, J.D.  
Samuel Huron  
Shneiderman, B.  
Liu, Z.  
Patterson, R.E.  
Stasko, J.T.  
Wiederkehr, B.  
Designation Agentur München.  
Ware, C.  
Ware, C.  
EMAPS project.

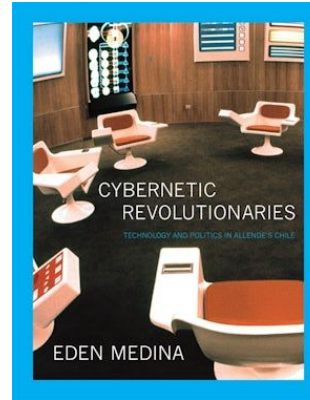
## Search for possible new documents

### Prototyping chatbot to build questions system & issues about the project.

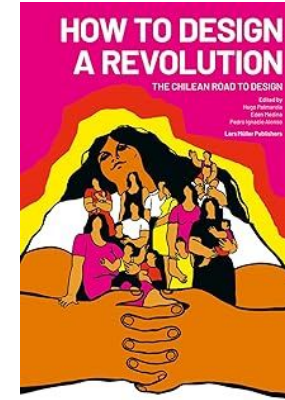
To give context & keep independence form SBA



*Proyecto Synco, Conceptos y práctica del control; una experiencia concreta: La dirección industrial en Chile. 1973. Santiago. 65 p.*



Eden Medina Book?

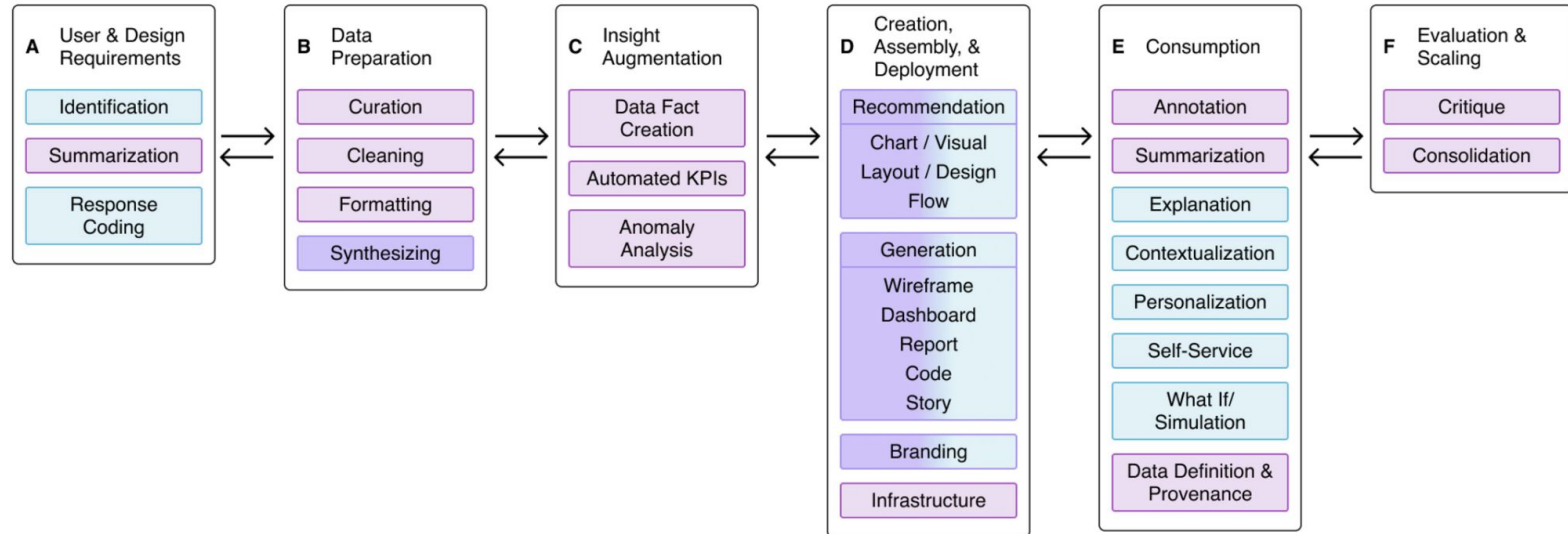


Recent Exhibition Book?

## Diagram and Tasks

### Visualization Workflow Phases

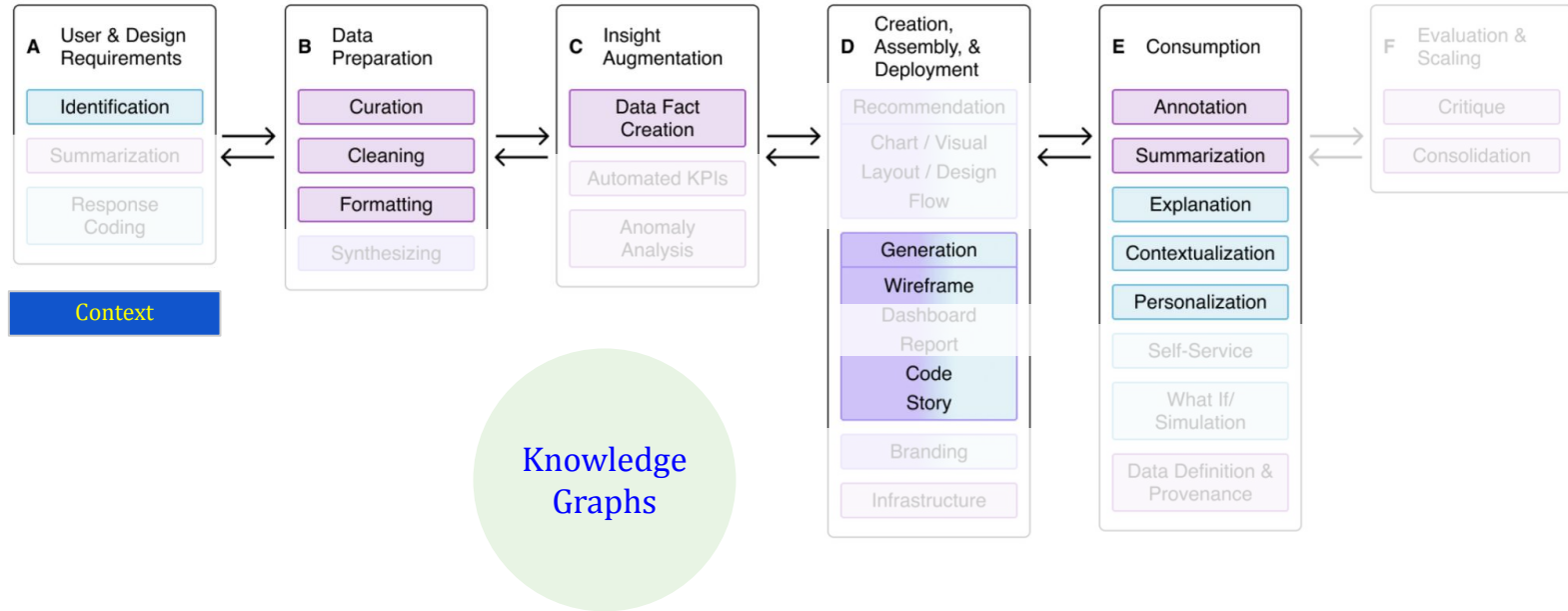
Types of Generative AI: ■ Creativity ■ Assistance ■ Automation



## Diagram and Tasks: Priorities

### Visualization Workflow Phases

Types of Generative AI: ■ Creativity ■ Assistance ■ Automation



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